

Algorithmic Game Theory

1st Problem Set

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Problem 1

Solution.

We examine each statement separately.

1. **False.** Strategy B is not strictly dominated by A , since against Z we have

$$u_1(B, Z) = 1 > 0 = u_1(A, Z).$$

2. **True.** For player 2, column X strictly dominates column Y , since

$$40 > 23, \quad 14 > 13, \quad 5 > 3, \quad 11 > 5.$$

3. **False.** Strategy B is not weakly dominated by D , since against Y ,

$$u_1(B, Y) = 5 > 3 = u_1(D, Y).$$

4. **True.** Strategy C weakly dominates A , because it strictly dominates A , since

$$20 > 15, \quad 20 > 13, \quad 11 > 9, \quad 1 > 0.$$

5. **False.** Column Z is not strictly dominated by Y , because against row A ,

$$u_2(A, Y) = 23 = 23 = u_2(A, Z).$$

So Y only weakly dominates Z .

6. **False.** Player 1 does have a dominant strategy: C . It gives at least as much payoff as every other row against every column.

7. **False.** Against row A , player 2's payoffs are

$$W : 12, \quad X : 40, \quad Y : 23, \quad Z : 23.$$

The best response is X , not Z .

8. **True.** Against column W , player 1's payoffs are

$$A : 15, \quad B : 2, \quad C : 20, \quad D : 20.$$

So C is a best response, tied with D .

9. **True.** The pure Nash equilibria are (C, W) and (D, W) . At (C, W) , the social welfare is

$$20 + 7 = 27 < 30.$$

Problem 2

Solution.

(i)

We are given the following zero-sum game, where the entries denote the payoff of the row player:

	t_1	t_2
s_1	a	7
s_2	10	d

where

$$a > 10, \quad 0 \leq d < 10.$$

Since this is a zero-sum game, a pure Nash equilibrium is a strategy environment where:

- the row player cannot increase their payoff by changing strategy, and
- the column player cannot decrease the row player's payoff by changing strategy.

We first check what the row player would prefer in each column:

$$\min(a, 7) = 7,$$

and

$$\min(10, d) = d.$$

Therefore,

$$\max\{7, d\}$$

is the row player's maximin value.

Now we compute the column player's choices. In column t_1 , the maximum payoff of the row player is

$$\max(a, 10) = a.$$

In column t_2 , the maximum payoff of the row player is

$$\max(7, d).$$

Since $a > 10$ and $\max(7, d) < 10$, the column player prefers column t_2 .

From the above, we see that:

- If $d < 7$, then (s_1, t_2) is the unique pure Nash equilibrium.
- If $d = 7$, then both (s_1, t_2) and (s_2, t_2) are pure Nash equilibria.
- If $d > 7$, then (s_2, t_2) is the unique pure Nash equilibrium.

Therefore, for all values of $a > 10$ and $0 \leq d < 10$, a pure Nash equilibrium always exists.

(ii)

Each player has four pure strategies:

$$A1-B1, \quad A1-B2, \quad A2-B1, \quad A2-B2.$$

Here, A_i means that the player shows i fingers, while B_j means that the player guesses that the opponent shows j fingers.

The payoff rule is the following:

- if both players guess correctly, the payoff is 0
- if both players guess incorrectly, the payoff is 0
- if only the row player guesses correctly, the row player receives the sum of the shown fingers
- if only the column player guesses correctly, the row player loses the sum of the shown fingers

Using the order

$$A1-B1, \quad A1-B2, \quad A2-B1, \quad A2-B2,$$

the zero-sum payoff matrix for the row player is:

	$A1-B1$	$A1-B2$	$A2-B1$	$A2-B2$
$A1-B1$	0	2	-3	0
$A1-B2$	-2	0	0	3
$A2-B1$	3	0	0	-4
$A2-B2$	0	-3	4	0

Now we examine whether there exists an equilibrium where both players use mixed strategies with support only

$$\{A1-B2, A2-B1\}.$$

Let the row player play $A1-B2$ with probability p and $A2-B1$ with probability $1 - p$.

Let the column player play $A1-B2$ with probability q and $A2-B1$ with probability $1 - q$.

Inside this support, all payoffs are zero:

	$A1-B2$	$A2-B1$
$A1-B2$	0	0
$A2-B1$	0	0

So the value of the game in this support is 0.

We must check that neither player wants to deviate outside the support.
For the row player, the expected payoff from deviating to $A1-B1$ is

$$2q - 3(1 - q) = 5q - 3.$$

Thus, we need

$$5q - 3 \leq 0 \iff q \leq \frac{3}{5}.$$

The expected payoff from deviating to $A2-B2$ is

$$-3q + 4(1 - q) = 4 - 7q.$$

Thus, we need

$$4 - 7q \leq 0 \iff q \geq \frac{4}{7}.$$

So we need

$$\frac{4}{7} \leq q \leq \frac{3}{5}.$$

Similarly, for the column player, we require that no deviation gives the row player payoff less than 0.

If the column player deviates to $A1-B1$, the row player's expected payoff is

$$-2p + 3(1 - p) = 3 - 5p.$$

We need

$$3 - 5p \geq 0 \iff p \leq \frac{3}{5}.$$

If the column player deviates to $A2-B2$, the row player's expected payoff is

$$3p - 4(1 - p) = 7p - 4.$$

We need

$$7p - 4 \geq 0 \iff p \geq \frac{4}{7}.$$

Therefore, we need

$$\frac{4}{7} \leq p \leq \frac{3}{5}.$$

Hence, such a mixed Nash equilibrium exists.

Therefore, both players can mix only between $A1-B2$ and $A2-B1$ in equilibrium.

Problem 3

Solution.

(i)

We choose

$$x = 2.$$

Then the game becomes

	t_1	t_2
s_1	(4, 4)	(2, 4)
s_2	(4, 2)	(4, 5)

We now find the Nash equilibria.

For player 1:

- If player 2 plays t_1 , then player 1 gets 4 from both s_1 and s_2 .
- If player 2 plays t_2 , then player 1 prefers s_2 , since $4 > 2$.

For player 2:

- If player 1 plays s_1 , then player 2 gets 4 from both t_1 and t_2 .
- If player 1 plays s_2 , then player 2 prefers t_2 , since $5 > 2$.

Therefore, the pure Nash equilibria are

$$(s_1, t_1) \quad \text{and} \quad (s_2, t_2).$$

There is no mixed Nash equilibrium where both players mix with positive probability. Hence, for $x = 2$, all Nash equilibria are

$$(s_1, t_1) \quad \text{and} \quad (s_2, t_2).$$

(ii)

The game is

	t_1	t_2	t_3	t_4	t_5
s_1	(1, 0)	(4, 2)	(5, 2)	(3, 4)	(6, 5)
s_2	(2, 6)	(3, 5)	(4, 3)	(4, 3)	(1, 0)

First, we find the pure Nash equilibria.

Player 1's best responses are:

	t_1	t_2	t_3	t_4	t_5
Best response	s_2	s_1	s_1	s_2	s_1

Player 2's best responses are:

	s_1	s_2
Best response	t_5	t_1

Therefore, the pure Nash equilibria are

$$(s_1, t_5) \quad \text{and} \quad (s_2, t_1).$$

Now we look for mixed Nash equilibria.

Let player 1 play s_1 with probability p and s_2 with probability $1 - p$.

Then player 2's expected payoffs from the columns are:

$$\begin{aligned} u_2(t_1) &= 6 - 6p, & u_2(t_2) &= 5 - 3p, & u_2(t_3) &= 3 - p, \\ u_2(t_4) &= 3 + p, & u_2(t_5) &= 5p. \end{aligned}$$

By comparing these payoffs, player 2's best responses are:

$$\left\{ \begin{array}{ll} t_1, & p < \frac{1}{3}, \\ t_1, t_2, & p = \frac{1}{3}, \\ t_2, & \frac{1}{3} < p < \frac{1}{2}, \\ t_2, t_4, & p = \frac{1}{2}, \\ t_4, & \frac{1}{2} < p < \frac{3}{4}, \\ t_4, t_5, & p = \frac{3}{4}, \\ t_5, & p > \frac{3}{4}. \end{array} \right.$$

For player 1 to mix, player 1 must be indifferent between s_1 and s_2 .

Let player 2 use probabilities

$$(q_1, q_2, q_3, q_4, q_5)$$

over

$$t_1, t_2, t_3, t_4, t_5.$$

Player 1's payoff from s_1 is

$$q_1 + 4q_2 + 5q_3 + 3q_4 + 6q_5,$$

and from s_2 is

$$2q_1 + 3q_2 + 4q_3 + 4q_4 + q_5.$$

Indifference requires:

$$q_1 + 4q_2 + 5q_3 + 3q_4 + 6q_5 = 2q_1 + 3q_2 + 4q_3 + 4q_4 + q_5,$$

that is,

$$-q_1 + q_2 + q_3 - q_4 + 5q_5 = 0.$$

We now examine the relevant cases.

Case 1: $p = \frac{1}{3}$.

Player 2 mixes among t_1, t_2 , so

$$q_3 = q_4 = q_5 = 0, \quad q_1 + q_2 = 1.$$

The indifference condition becomes

$$-q_1 + q_2 = 0 \Rightarrow q_1 = q_2 = \frac{1}{2}.$$

Hence, a mixed Nash equilibrium is:

$$p = \frac{1}{3}, \quad (q_1, q_2, q_3, q_4, q_5) = \left(\frac{1}{2}, \frac{1}{2}, 0, 0, 0\right).$$

Case 2: $p = \frac{1}{2}$.

Player 2 mixes among t_2, t_4 , so

$$q_1 = q_3 = q_5 = 0, \quad q_2 + q_4 = 1.$$

The indifference condition becomes

$$q_2 - q_4 = 0 \Rightarrow q_2 = q_4 = \frac{1}{2}.$$

Thus, another mixed Nash equilibrium is:

$$p = \frac{1}{2}, \quad (q_1, q_2, q_3, q_4, q_5) = \left(0, \frac{1}{2}, 0, \frac{1}{2}, 0\right).$$

Case 3: $p = \frac{3}{4}$.

Player 2 mixes among t_4, t_5 , so

$$q_1 = q_2 = q_3 = 0, \quad q_4 + q_5 = 1.$$

The indifference condition becomes

$$-q_4 + 5q_5 = 0.$$

Solving together with $q_4 + q_5 = 1$, we get:

$$q_5 = \frac{1}{6}, \quad q_4 = \frac{5}{6}.$$

Hence, another mixed Nash equilibrium is:

$$p = \frac{3}{4}, \quad (q_1, q_2, q_3, q_4, q_5) = \left(0, 0, 0, \frac{5}{6}, \frac{1}{6}\right).$$

Therefore, all Nash equilibria are:

$$(s_1, t_5), \quad (s_2, t_1),$$

and the mixed equilibria:

$$\begin{aligned} p &= \frac{1}{3}, & \left(\frac{1}{2}, \frac{1}{2}, 0, 0, 0\right), \\ p &= \frac{1}{2}, & \left(0, \frac{1}{2}, 0, \frac{1}{2}, 0\right), \\ p &= \frac{3}{4}, & \left(0, 0, 0, \frac{5}{6}, \frac{1}{6}\right). \end{aligned}$$

Problem 4

Solution.

(i)

We model the voting process as a normal-form game.

Players: There are 50 players, one for each student.

Strategies: Each player chooses one party to vote for. Thus, for every player i ,

$$S_i = \{\Pi_1, \Pi_2, \Pi_3, \Pi_4\}.$$

Outcome: Given a strategy profile, we count the votes for each party. The party with the most votes wins. In case of a tie, the party with the smallest index wins.

Utility functions: Each player has a strict preference ordering over the parties, as given in Table 1.

We define the utility of each player depending only on the winning party. For example, if player i 's preferences are:

$$\Pi_a \succ \Pi_b \succ \Pi_c \succ \Pi_d,$$

then we can define the utility function u_i as:

$$u_i(\Pi_a) = 4, \quad u_i(\Pi_b) = 3, \quad u_i(\Pi_c) = 2, \quad u_i(\Pi_d) = 1.$$

Thus, given any strategy profile, each player receives a utility equal to the value assigned to the winning party according to their preferences.

(ii)

We show that for every $i \in \{1, 2, 3, 4\}$, there exists a Nash equilibrium in which party Π_i wins.

We fix some party Π_i . Let's consider the strategy profile where all 50 players vote for Π_i .

Then Π_i receives 50 votes, while all other parties receive 0 votes. Hence, Π_i is the winner.

Now consider any player. Suppose they deviate and vote for a different party.

Then Π_i receives 49 votes, while the other party receives 1 vote. Therefore, Π_i still wins.

Thus, a single player cannot change the outcome of the election by deviating. Since the outcome does not change, the player's utility also does not change.

Hence, no player has an incentive to deviate.

Therefore, this strategy profile is a Nash equilibrium.

Since this argument holds for any party Π_i , we conclude that for every $i \in \{1, 2, 3, 4\}$, there exists a Nash equilibrium in which Π_i wins.

(iii)

From Table 1, the vote counts are

$$(\Pi_1, \Pi_2, \Pi_3, \Pi_4) = (20, 10, 20, 0).$$

Thus, Π_1 wins, because Π_1 and Π_3 are tied and ties are broken in favour of the party with the smaller index.

We want a Nash equilibrium in which Π_2 wins, while minimizing the number of voters who do not vote for their true first preference.

Suppose that:

$$x$$

voters whose first preference is Π_1 vote for Π_2 , and

$$y$$

voters whose first preference is Π_3 vote for Π_2 .

Then the new vote counts are:

$$\Pi_1 = 20 - x,$$

$$\Pi_2 = 10 + x + y,$$

$$\Pi_3 = 20 - y.$$

Since ties between Π_1 and Π_2 are won by Π_1 , party Π_2 must remain ahead even if one voter changes from Π_2 to Π_1 . Therefore, we need

$$(10 + x + y) - 1 > (20 - x) + 1.$$

Equivalently,

$$10 + x + y > 20 - x + 2,$$

so

$$2x + y > 12.$$

Since x, y are integers, this is equivalent to

$$2x + y \geq 13.$$

Also, Π_2 must beat Π_3 . Since Π_2 has a smaller index than Π_3 , a tie between them would be won by Π_2 . However, in order to be stable against one voter deviating from Π_2 to Π_3 , we need Π_2 to have at least one more vote than Π_3 . Thus:

$$(10 + x + y) - 1 \geq (20 - y) + 1.$$

Equivalently,

$$x + 2y \geq 12.$$

Therefore, we need:

$$2x + y \geq 13$$

and

$$x + 2y \geq 12.$$

We want to minimize:

$$x + y.$$

If $x + y = 8$, then the first inequality gives

$$2x + y = x + 8 \geq 13,$$

so

$$x \geq 5.$$

The second inequality gives

$$x + 2y = 8 + y \geq 12,$$

so

$$y \geq 4.$$

But then

$$x + y \geq 9,$$

which contradicts $x + y = 8$. Hence at least 9 voters must vote differently from their true first preference.

Now take:

$$x = 4, \quad y = 5.$$

Indeed,

$$2x + y = 2 \cdot 4 + 5 = 13,$$

and

$$x + 2y = 4 + 2 \cdot 5 = 14.$$

Then:

$$x + y = 9.$$

The vote counts become:

$$\Pi_1 = 20 - 4 = 16,$$

$$\Pi_2 = 10 + 4 + 5 = 19,$$

$$\Pi_3 = 20 - 5 = 15,$$

$$\Pi_4 = 0.$$

So:

$$(\Pi_1, \Pi_2, \Pi_3, \Pi_4) = (16, 19, 15, 0).$$

Hence, Π_2 wins.

Moreover, no single voter can change the winner by deviating:

- If one voter changes to Π_1 , then Π_1 can get at most 17 votes, while Π_2 still has at least 18 votes.
- If one voter changes to Π_3 , then Π_3 can get at most 16 votes, while Π_2 still has at least 18 votes.
- A deviation to Π_4 also cannot make Π_4 win.

Therefore, no voter can unilaterally change the winner to a party they prefer more than Π_2 . Hence this voting profile is a Nash equilibrium.

Thus, the minimum number of voters who must not vote for their true first preference is 9.

Problem 5

Solution.

Consider any pure Nash equilibrium

$$(b_1, b_2, \dots, b_n).$$

Let

$$B = \max_i b_i$$

be the highest bid.

We prove the two required statements.

1. The two highest bids are equal, and one of them is submitted by player 1.

First, we show that player 1 must be the winner.

Suppose, for contradiction, that some player $j \neq 1$ wins with bid b_j . Since player j wins, their utility is

$$v_j - b_j.$$

In equilibrium, this utility cannot be negative, because player j could instead bid 0 and get utility at least 0. Hence,

$$b_j \leq v_j.$$

Since

$$v_1 > v_j,$$

we get

$$v_1 - b_j > 0.$$

Then player 1 could deviate and bid b_j . Since player 1 has the smallest index, player 1 would win any tie at bid b_j . This would give player 1 positive utility:

$$v_1 - b_j > 0.$$

This is a profitable deviation, contradicting equilibrium.
Therefore, player 1 must be the winner.

Now suppose that player 1's winning bid is strictly larger than every other bid. Let

$$M = \max_{i \neq 1} b_i$$

be the second highest bid. Suppose

$$b_1 > M.$$

Then player 1 can slightly lower their bid to some bid b'_1 such that

$$M < b'_1 < b_1.$$

Player 1 still wins, because their bid is still higher than every other bid. However, player 1 pays less, so their utility increases from

$$v_1 - b_1$$

to

$$v_1 - b'_1.$$

Since

$$b'_1 < b_1,$$

we have

$$v_1 - b'_1 > v_1 - b_1.$$

This is a profitable deviation, contradicting equilibrium.

Therefore, player 1's bid cannot be strictly above the second highest bid. Hence, the highest bid must be tied with at least one other bid.

So the two highest bids are equal, and one of them is submitted by player 1.

Thus, $b_1 = B$ and at least one other player also bids B .

2. The two highest bids belong to the interval $[v_2, v_1]$.

We now show that

$$B \in [v_2, v_1].$$

First, since player 1 wins and pays B , their utility is

$$v_1 - B.$$

In equilibrium, player 1 cannot receive negative utility, because they could bid 0 instead and get utility at least 0. Therefore,

$$v_1 - B \geq 0.$$

Hence,

$$B \leq v_1.$$

Now we show that

$$B \geq v_2.$$

Suppose, for contradiction, that

$$B < v_2.$$

Then player 2 could deviate and bid some amount

$$B + \varepsilon$$

where

$$0 < \varepsilon < v_2 - B.$$

Then player 2 would become the winner, because their bid would be strictly higher than all previous bids.

Their utility would be

$$v_2 - (B + \varepsilon).$$

Since

$$B + \varepsilon < v_2,$$

this utility is positive:

$$v_2 - (B + \varepsilon) > 0.$$

Before deviating, player 2 was not winning against player 1 at the tied highest bid, because ties are broken in favour of the smaller index. Thus, player 2's utility was 0.

So player 2 has a profitable deviation, contradicting equilibrium.

Therefore,

$$B \geq v_2.$$

Combining the two inequalities, we get

$$v_2 \leq B \leq v_1.$$

Thus, in any pure Nash equilibrium:

1. the two highest bids are equal, and one of them is submitted by player 1;
2. these two highest bids belong to the interval

$$[v_2, v_1].$$

Problem 6

Solution.

Let's consider the following bid environment:

$$b_1 = 0, \quad b_2 = v_1, \quad b_i = 0 \text{ for every } i \geq 3.$$

Then player 2 submits the highest bid, since

$$b_2 = v_1 > 0.$$

Therefore, player 2 wins the auction.

Since this is a second-price auction, player 2 pays the second-highest bid, which is 0. Hence player 2's utility is:

$$u_2 = v_2 - 0 = v_2 > 0.$$

Now we check that no player has a profitable deviation.

Player 1 currently loses and gets utility 0. If player 1 wants to win, they must bid at least v_1 . If they bid exactly v_1 , then player 1 wins by the tie-breaking rule, but pays v_1 , so their utility is

$$v_1 - v_1 = 0.$$

If they bid more than v_1 , they still pay v_1 , so their utility is again

$$v_1 - v_1 = 0.$$

Thus, player 1 cannot get utility strictly greater than 0 by deviating.

Player 2 is winning and gets utility v_2 . If player 2 lowers their bid below or equal to 0, they may lose and get utility 0, which is worse.

If they keep winning, the price remains 0, so their utility does not improve.

Finally, any player $i \geq 3$ has valuation

$$v_i < v_2 < v_1.$$

To beat player 2, such a player would need to bid more than v_1 . Then they would win and pay v_1 , giving utility

$$v_i - v_1 < 0.$$

So no such player has a profitable deviation.

Therefore, this bid environment is a pure Nash equilibrium.

In this equilibrium, the winner is player 2, not player 1, even though player 1 has the highest valuation.

$$(b_1, b_2, b_3, \dots, b_n) = (0, v_1, 0, \dots, 0)$$